

CS 255 System Design Document

DriverPass System

This completed system design document replaces the template placeholders with complete DriverPass design content. It includes all required UML diagrams, formal UML relationship markings, detailed diagram explanations, and expanded technical requirements aligned to the client needs from the DriverPass interview.

Executive Summary

DriverPass needs a secure web-based system that helps customers prepare for DMV driving tests through online training, practice tests, and on-the-road lesson scheduling. The design supports customers, secretaries, the owner, the IT administrator, and DMV content updates. The solution provides cloud access from computers and mobile devices, role-based access, audit tracking, reporting, reservation management, and secure handling of customer information.

Client Needs and Design Response

Client Need	Design Response
Customers can schedule, modify, or cancel lessons online.	Use case, activity, sequence, and reservation classes support online reservation management.
Secretary can schedule lessons for customers who call or visit.	Secretary is included as a system user who can enter student information and manage reservations.
Each lesson must assign a driver and car.	Scheduling use case includes driver/car assignment; class diagram ties each reservation to one driver and one car.
Different users need different permissions.	UserAccount role/status fields support customer, secretary, owner, and IT administrator permissions.
Activity must be traceable.	AuditLog class and <<include>> relationships record who creates, changes, or cancels important records.
DMV rules and practice questions must stay current.	DMV System actor and DMVUpdate class feed updated rules and questions into practice testing.
Reports must be downloadable for review.	Owner and IT administrator can generate activity reports based on system data.
System must be accessible from computers and mobile devices.	Cloud-hosted browser-based application supports modern desktop and mobile browsers.

UML Use Case Diagram

The use case diagram exists to show what each major user group needs to accomplish in the DriverPass system. Customers can register, log in, purchase a package, schedule or modify lessons, take online classes and practice tests, and view progress. The secretary can enter student information and manage lesson reservations for customers who call or visit the office. The owner can disable packages and generate reports, while the IT administrator manages accounts and access. The DMV system provides rule and question updates. Formal `<<include>>` markings show required supporting behavior, such as scheduling including driver/car assignment and audit tracking. The `<<extend>>` relationship shows password reset as an optional extension of logging in when the user cannot access the account.

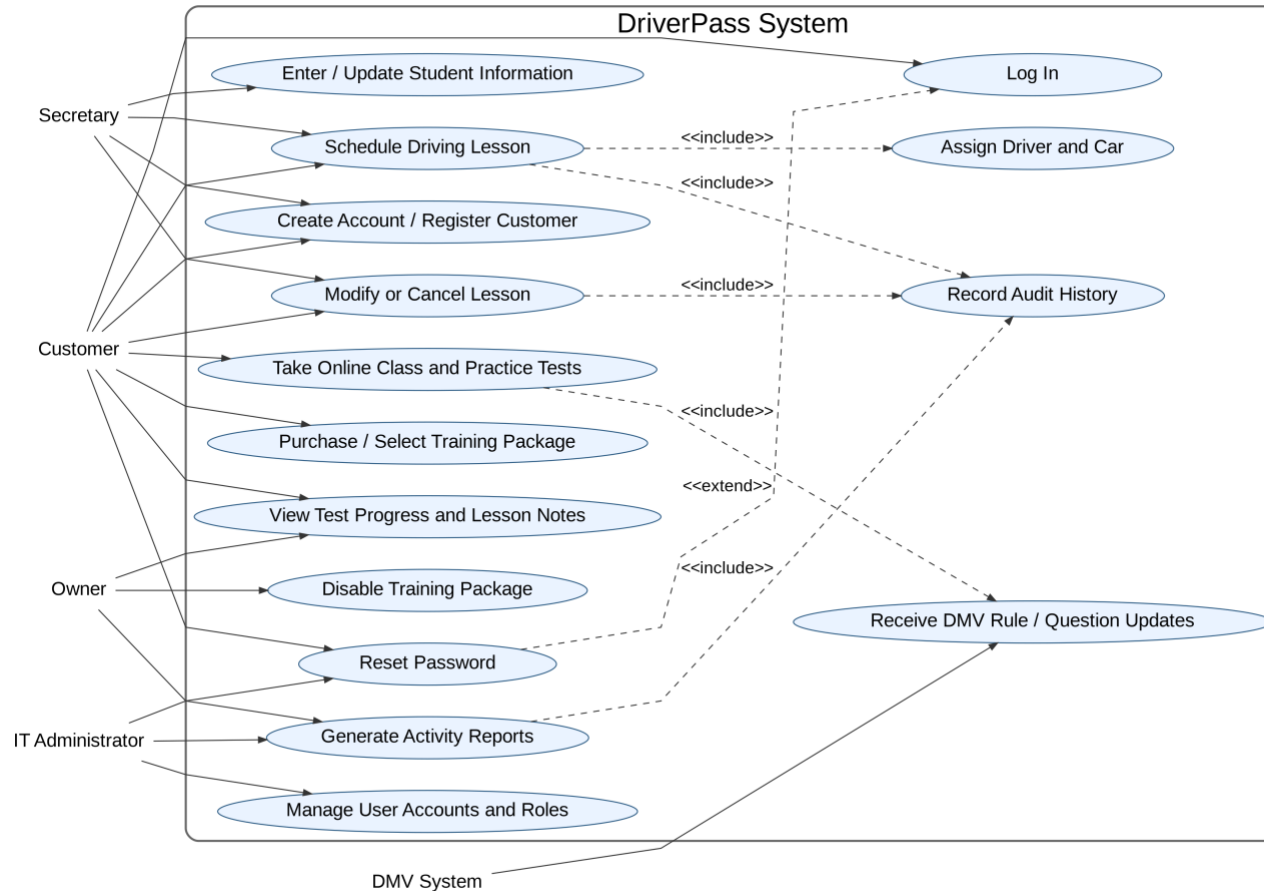


Figure 1. DriverPass use case diagram with formal `<<include>>` and `<<extend>>` relationships.

This use case diagram shows how DriverPass users interact with the system. Customers can create accounts, log in, select training packages, schedule or modify driving lessons, take online classes, complete practice tests, and view lesson notes. The secretary can enter customer information and schedule lessons for customers who call or visit the office. The owner can generate activity reports and disable training packages when needed. The IT administrator manages user accounts, resets access, and controls employee permissions. The DMV system provides rule and question updates so DriverPass training content stays current.

The diagram uses <<include>> relationships where one action is required as part of another action. For example, scheduling a driving lesson includes assigning a driver and car because every appointment must be matched with a specific instructor and vehicle. Taking a practice test includes loading current DMV-based questions so the test content remains accurate. The diagram also uses an <<extend>> relationship for password reset because resetting a password only occurs when a user cannot log in normally. These relationships make the design more complete and show how the system supports DriverPass's daily business needs.

UML Activity Diagram 1 - Schedule Driving Lesson

This activity diagram breaks down the scheduling process. It begins when a customer or secretary logs in, selects the customer and package, and enters the preferred date, time, pickup location, and drop-off location. The system checks package hours, driver availability, car availability, and schedule conflicts before confirming the reservation. If no slot is available, the user is directed to choose another time. If the slot is available, the system reserves the driver and car, saves the appointment, records the audit history, and sends confirmation. This directly supports DriverPass requirements for lesson scheduling, resource assignment, and accountability.

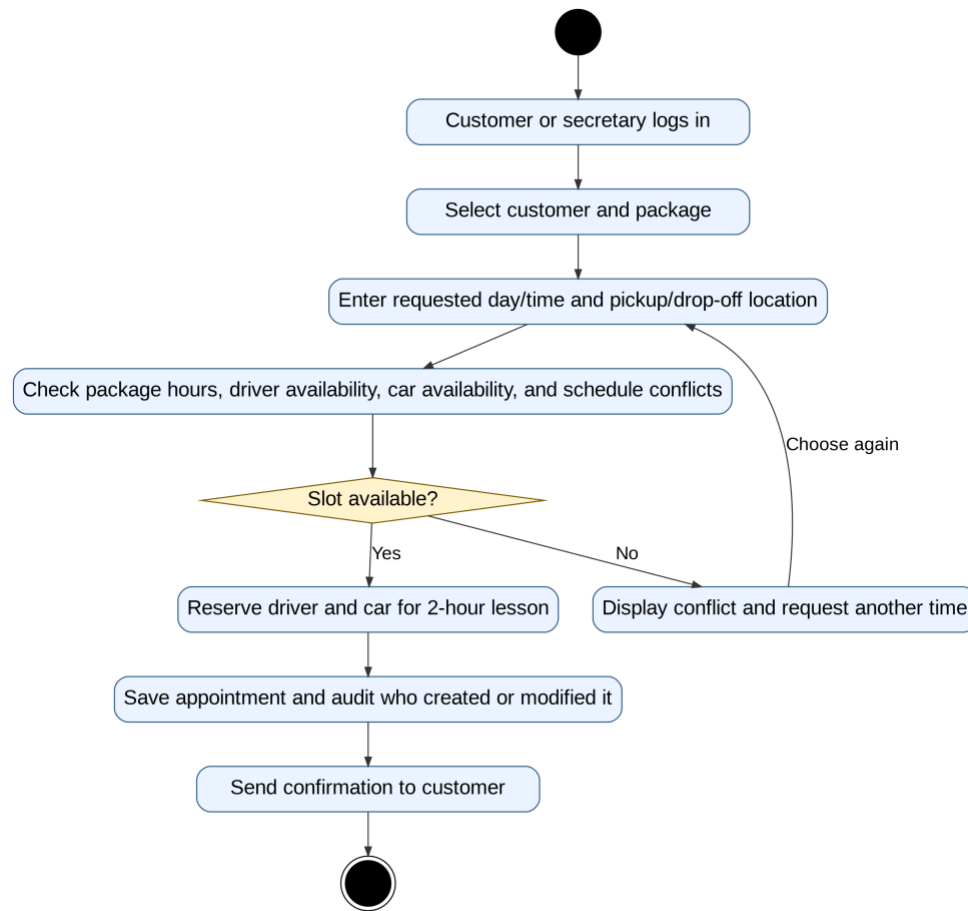


Figure 2. Activity diagram for scheduling a two-hour driving lesson.

This activity diagram explains the complete workflow for scheduling a two-hour driving lesson. The process begins when either a customer or secretary logs into the system and selects the customer and package. The system then collects the requested date, time, pickup location, and drop-off location. Before confirming the appointment, the system checks whether the customer has enough package hours remaining and whether a driver, car, and time slot are available.

This workflow supports DriverPass's need to prevent scheduling conflicts. If a slot is unavailable, the system displays the conflict and asks the user to choose another time. If the slot is available, the system reserves the driver and car, saves the appointment, records who created or changed the reservation, and sends confirmation to the customer. This directly supports the client's requirement to track who created, modified, or canceled appointments.

UML Activity Diagram 2 - Take Practice Test

This activity diagram shows how online learning is controlled by package access. After the customer logs in, the system checks whether the selected package includes online class access. If the package does not include it, the system displays a restriction or upgrade message. If access is allowed, the customer selects the online class or practice test, receives current DMV-based questions, answers the questions, submits the test, and receives a graded result. The system records time taken and updates the customer status as not taken, in progress, failed, or passed. This supports DriverPass customer progress tracking and DMV-aligned practice testing.

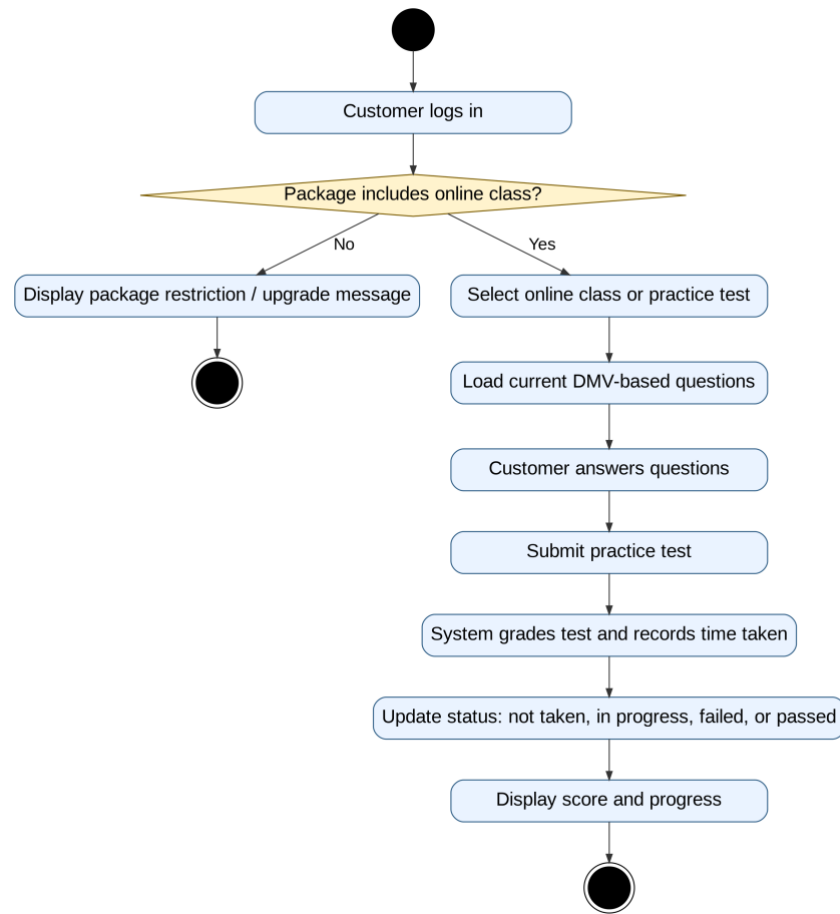


Figure 3. Activity diagram for taking an online practice test.

This activity diagram shows how a customer completes an online practice test. The customer logs in, and the system checks whether the selected training package includes online class access. If the package does not include online access, the system displays a restriction or upgrade message. If access is allowed, the customer selects an online class or practice test, and the system loads current DMV-based questions.

This workflow supports DriverPass's goal of helping students prepare for the DMV exam. The system records the customer's answers, grades the test, saves the score and time taken, and updates the test status as not taken, in progress, failed, or passed. This matches the client's request for online test progress tracking and ensures students can clearly see their preparation results.

UML Sequence Diagram - Schedule Driving Lesson

The sequence diagram shows the order of communication between the customer, web portal, scheduling service, database, and notification service. The customer requests a lesson through the web portal. The scheduling service checks package hours, driver availability, vehicle availability, and conflicts against the database. The system returns available options or a conflict message. Once the customer confirms, the reservation, assigned driver, assigned car, and audit record are stored, and the notification service sends confirmation. This diagram fixes the readability issue by using large labels and a clear numbered order.

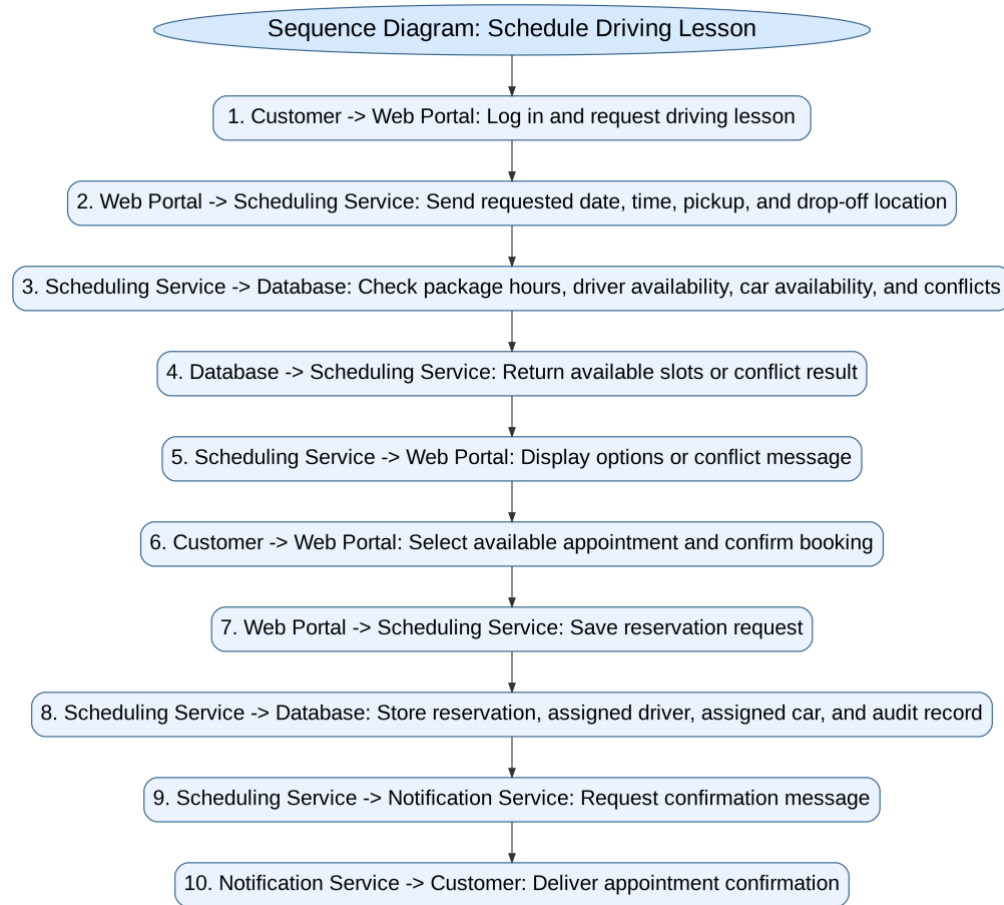


Figure 4. Readable sequence diagram for the driving lesson scheduling interaction.

This sequence diagram shows the order of communication between the customer, web portal, scheduling service, database, and notification service during lesson scheduling. The customer begins by logging in and requesting a lesson time. The web portal sends the request to the scheduling service, which checks package hours, driver availability, car availability, and appointment conflicts through the database.

If an appointment is available, the system saves the reservation and audit record, then sends a confirmation request to the notification service. The notification service delivers the confirmation to the customer. This diagram is important because it shows how the system components work together behind the scenes to complete a scheduling request accurately and securely.

UML Class Diagram

The class diagram defines the main data objects needed to build DriverPass. The Customer class stores personal and location information. UserAccount supports login, password reset, role, and status. TrainingPackage stores package name, hours, price, online access, and active status. Reservation ties a customer to a scheduled lesson and connects each reservation to exactly one driver and one car. PracticeTest stores DMV-based test progress, score, and time taken. DriverNote captures lesson feedback, AuditLog records system activity, and DMVUpdate supports updated DMV content. The multiplicities clarify how many records can be associated with each class, strengthening the design for developers and database planning.

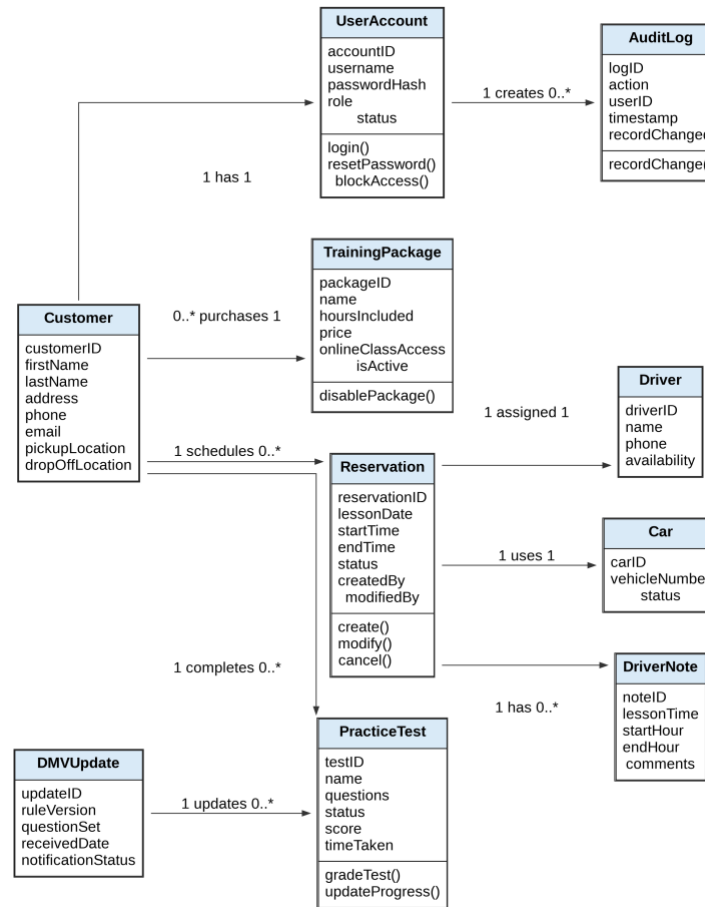


Figure 5. DriverPass class diagram with clear relationships and multiplicities.

This class diagram identifies the major objects needed to support DriverPass operations. The Customer class stores student information such as name, address, phone number, email, pickup location, and drop-off location. The UserAccount class supports login, password reset, user roles, and account status. The TrainingPackage class stores package information, pricing, hours included, online access, and whether the package is active. The Reservation class stores lesson date, start time, end time, status, and audit information.

The relationships show how the system data connects. One customer has one user account. One customer can purchase one or more training packages and schedule zero or more reservations. Each reservation must be assigned to one driver and one car. A customer can complete zero or more practice tests. Driver notes are connected to reservations so the system can record lesson times and instructor comments. Audit logs record important changes so DriverPass can track who created, modified, or canceled records.

Technical Requirements

Hardware Requirements

The DriverPass system should run on cloud-hosted servers so the company does not need to maintain physical servers in the office. Employees should be able to access the system using standard office computers, laptops, and tablets. Customers should be able to access the system from personal computers, smartphones, and tablets. DriverPass will also need reliable internet access at the office so the secretary, owner, and IT administrator can use the system during business operations.

Software Requirements

The system should be developed as a web-based application that works through modern browsers such as Chrome, Edge, Safari, and Firefox. The application should include customer registration, login, password reset, package selection, lesson scheduling, practice testing, reporting, and administrative account management. The system should also support downloadable reports that can be opened in programs such as Microsoft Excel.

Database Requirements

The system should use a relational database to store customer records, user accounts, training packages, lesson reservations, drivers, cars, practice test results, driver notes, DMV updates, and audit logs. The database should support relationships between customers, reservations, drivers, vehicles, and practice tests so the system can prevent scheduling conflicts and maintain accurate records.

Cloud and Infrastructure Requirements

The system should be hosted in a secure cloud environment. Cloud hosting supports DriverPass's need for access from multiple computers and mobile devices while reducing the company's responsibility for server maintenance, backups, and security management. The cloud environment should support system monitoring, automatic updates, storage, backup services, and recovery in case of outage or data loss.

Security Requirements

The system should use HTTPS/TLS encryption to protect data during transmission. User access should be controlled by roles so customers, secretaries, owners, and IT administrators only have access to the functions they need. Passwords should be stored securely using hashing, and the system should provide password reset functionality. The IT administrator should be able to block or disable accounts when employees leave the company. Audit logs should track who creates, modifies, or cancels reservations and other important records.

Payment Security Requirements

Because DriverPass collects payment information, credit card data should be handled through a secure payment provider instead of being stored directly in the DriverPass database. This reduces risk and helps protect customer financial information.

Backup and Disaster Recovery Requirements

The system should include automatic backups of database records and application files. Backup copies should be stored securely and tested regularly to make sure the system can be restored after a technical failure, accidental deletion, or cloud service outage. Recovery procedures should allow DriverPass to continue operations with minimal downtime.

Scalability and Availability Requirements

The system should be designed to support future growth. As DriverPass adds more customers, drivers, cars, packages, or online training content, the cloud infrastructure should be able to scale without requiring a complete redesign. The system should be available to customers and employees whenever internet access is available, especially because customers may schedule lessons outside normal office hours.

Reporting Requirements

The system should allow the owner and authorized staff to generate activity reports showing who created, modified, or canceled reservations. Reports should also support customer progress tracking, package usage, lesson scheduling, and practice test performance. Reports should be downloadable so DriverPass can review business information offline when needed.

Final Design Summary

The final DriverPass design meets the client needs by combining secure account access, online training, practice testing, lesson scheduling, driver and vehicle assignment, reporting, DMV content updates, and audit tracking in one cloud-based system. The UML diagrams show how users interact with the system, how key workflows operate, how scheduling messages move through the system, and how data is organized. The technical requirements provide the hardware, software, infrastructure, database, security, and recovery details needed to build the system successfully and support future business growth.

The DriverPass system design meets the client's needs by supporting online customer registration, training package selection, lesson scheduling, driver and vehicle assignment, practice testing, DMV content updates, reporting, and administrative account management. The design also supports security through role-based access, encrypted communication, password reset, account blocking, payment protection, and audit tracking. Because the system is cloud-based, customers and employees can access it from computers and mobile devices while DriverPass avoids the burden of managing local servers and backups. Overall, the proposed design provides a secure, flexible, and scalable solution that supports DriverPass's current business goals and future growth.

References

Southern New Hampshire University. (n.d.). CS 255 DriverPass interview transcript.

Southern New Hampshire University. (n.d.). CS 255 system design document template.